

Package 'RegCalib'

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Type: Package

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Title: Regression Calibration for Measurement Error Correction

License: GPL-3

Depends: R ($\geq 3.5.0$)

Imports: stats, dplyr, Matrix, matrixcalc, earth, survival

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URL: https://github.com/YOUR_USERNAME/RegCalib

BugReports: https://github.com/YOUR_USERNAME/RegCalib/issues

Description: Corrects for measurement error in continuous exposures (and covariates) using regression calibration methods. Provides corrected coefficients, standard errors, p-values, and variance-covariance matrices for linear, generalized linear, and Cox proportional hazards outcome models under both external and internal validation study designs. Implements the Rosner-Spiegelman-Willett (RSW) delta-method approach (RegCalibDF) and the Carroll-Ruppert-Stefanski-Crainiceanu (CRS) sandwich-variance approach (RegCalibSub). Supports single and multiple mismeasured exposures.

Contents

RegCalibDF	2
RegCalibSub	5

RegCalibDF

Regression Calibration Using the Delta-Method (RSW Approach)

Description

Corrects for measurement error in continuous exposures (and covariates) and returns corrected coefficients, standard errors, p-values, and variance-covariance matrices. Users may supply their own uncorrected estimates (e.g., from Cox or logistic models) instead of using the built-in outcome model. Both external and internal validation study designs are supported.

Usage

```
RegCalibDF(  
  supplyEstimates = FALSE, ms, vs,  
  sur, exp, covCalib = NULL, covOutcome = NULL,  
  outcome = NA, event, time,  
  method = "lm", family = NA, link = NA,  
  external = TRUE,  
  pointEstimates = NA, vcovEstimates = NA  
)
```

Arguments

supplyEstimates

Logical. If TRUE, uncorrected estimates are supplied by the user via pointEstimates and vcovEstimates, and ms is optional. Standard regression results are not returned in this case. Default FALSE.

ms

Main study data frame. Must contain all variables specified in sur, covCalib, and covOutcome. Required when supplyEstimates = FALSE.

vs

Internal or external validation study data frame. Must contain all variables specified in exp, sur, and covCalib.

sur

Character vector of mismeasured exposure(s)/covariate(s) (surrogates) in the main study dataset.

exp

Character vector of correctly-measured exposure(s)/covariate(s) in the validation dataset. Must correspond one-to-one with sur and have the same length.

covCalib

Character vector of correctly-measured covariates to adjust for in both the calibration model and outcome model. Default NULL.

covOutcome

Character vector of correctly-measured risk factors for the outcome that are not associated with the exposure or surrogate. Included in outcome model only; must not overlap with covCalib. Default NULL.

outcome

Character. Name of the outcome variable. Required when method is "lm" or "glm".

event

Character. Name of the event status indicator for Cox models (0 = censored, 1 = event). Required when method = "cox".

time

Character. Name of the follow-up time variable for Cox models. Required when method = "cox".

method

Character. Outcome modelling method: "lm", "glm", or "cox". Default "lm".

family

Family function for glm (e.g., binomial). Not a character string. Required when method = "glm".

link

Character. Link function for glm (e.g., "logit"). Required when method = "glm".

external

Logical. TRUE (default) for an external validation study; FALSE for an internal validation study. When external = FALSE, vs must contain the outcome variable.

pointEstimates

Named numeric vector of uncorrected point estimates from standard regression (intercept excluded). Required when supplyEstimates = TRUE.

vcovEstimates

Named square matrix of uncorrected variance-covariance estimates (intercept excluded). Required when supplyEstimates = TRUE.

Value

A named list containing:

correctedCoefTable

Data frame of corrected estimates, standard errors, Z-values, p-values, and 95% confidence intervals.

correctedVCOV

Variance-covariance matrix of the corrected estimates.

standardCoefTable

(When supplyEstimates = FALSE) Results from the uncorrected standard regression.

standardVCOV

(When supplyEstimates = FALSE) Variance-covariance matrix from the uncorrected standard regression.

calibrationModelCoefTable

Calibration model slope estimates.

calibrationModelVCOV

Residual variance-covariance matrix from the calibration model.

Examples

```
result <- RegCalibDF(
  ms      = main_data,
```

```
vs      = validation_data,  
sur      = "Z",  
exp      = "X",  
covCalib = c("V1", "V2"),  
outcome  = "Y",  
method   = "lm",  
external = TRUE  
)  
result$correctedCoefTable  
result$correctedVCOV
```

References

Rosner B, Willett WC, Spiegelman D (1989). Correction of logistic relative risk estimates and confidence intervals for systematic within-person measurement error. *Statistics in Medicine* 8:1051-1069.

Rosner B, Spiegelman D, Willett WC (1990). Correction of logistic regression relative risk estimates and confidence intervals for measurement error: the case of multiple covariates measured with error. *American Journal of Epidemiology* 132:734-735.

Spiegelman D, McDermott A, Rosner B (1997). The many uses of the regression calibration method for measurement error bias correction in nutritional epidemiology. *American Journal of Clinical Nutrition* 65:1179S-1186S.

Spiegelman D, Carroll RJ, Kipnis V (2001). Efficient regression calibration for logistic regression in main study/internal validation study designs with an imperfect reference instrument. *Statistics in Medicine* 20:139-160.

See Also

RegCalibSub for the CRS sandwich variance approach.

RegCalibSub

Regression Calibration Using Sandwich Variance Estimation (CRS Approach)

Description

Corrects for measurement error in continuous exposures (and covariates) and returns corrected coefficients, standard errors, p-values, and variance-covariance matrices using the Carroll-Ruppert-Stefanski-Crainiceanu (CRS) sandwich variance estimator. Supports linear and generalized linear outcome models under external and internal validation study designs. Standard errors are derived analytically via the sandwich estimator (not bootstrap). Non-linear terms such as interaction terms should be pre-computed as permanent columns in the input datasets rather than specified in the formula.

Usage

```
RegCalibSub(  
  ms, vs, sur, exp, vsIndicator,  
  covCalib = NULL, covOutcome = NULL,  
  outcome = NA,  
  method = "lm", family = NA, link = NA,  
  external = TRUE  
)
```

Arguments

ms

Main study data frame. Must contain all variables specified in sur, outcome, and covOutcome (if any).

vs

External validation study data frame. Must contain all variables specified in exp, sur, and covCalib (if any). Required when external = TRUE.

sur

Character vector of mismeasured exposure(s)/covariate(s) (surrogates) in the main study dataset.

exp

Character vector of correctly-measured exposure(s)/covariate(s) in the validation dataset. Must correspond one-to-one with sur and have the same length.

vsIndicator

Character. Name of the indicator variable in the main study dataset identifying subjects with a validation record (1 = has validation record). Required when external = FALSE.

covCalib

Character vector of correctly-measured covariates to adjust for in the calibration model, including any non-linear terms. Default NULL.

covOutcome

Character vector of correctly-measured covariates to adjust for in the outcome model (with corrected exposure), including any non-linear terms. Default NULL.

outcome

Character. Name of the outcome variable. Required.

method

Character. Outcome modelling method: "lm" or "glm". Default "lm".

family

Family function for glm (e.g., binomial). Not a character string. Required when method = "glm".

link

Character. Link function for glm (e.g., "logit" or "log"). Required when method = "glm".

external

Logical. TRUE (default) for an external validation study; FALSE for an internal validation study embedded in ms.

Value

A named list containing:

correctedCoefTable

Matrix of corrected estimates, standard errors, Z-values, p-values, and 95% confidence intervals.

correctedVCOV

Variance-covariance matrix of the corrected estimates.

Examples

```
# External validation study, linear model
result <- RegCalibSub(
  ms      = main_data,
  vs      = validation_data,
  sur     = "Z",
  exp     = "X",
  covCalib = c("V1", "V2"),
  outcome = "Y",
  method  = "lm",
  external = TRUE
)
result$correctedCoefTable

# External validation study, logistic model
result <- RegCalibSub(
  ms      = main_data,
  vs      = validation_data,
  sur     = "Z",
  exp     = "X",
  covCalib = c("V1", "V2"),
  outcome = "Y",
  method  = "glm",
  family   = binomial,
  link     = "logit",
  external = TRUE
)
```

```
# Internal validation study
result <- RegCalibSub(
  ms      = main_data,
  sur      = "Z",
  exp      = "X",
  vsIndicator = "is_validated",
  covCalib  = c("V1", "V2"),
  outcome   = "Y",
  method    = "lm",
  external   = FALSE
)
```

References

Carroll RJ, Ruppert D, Stefanski LA, Crainiceanu CM (2006). *Measurement Error in Nonlinear Models*, 2nd ed. Chapman & Hall/CRC.

See Also

RegCalibDF for the RSW delta-method approach.